Practice workbook · Calculate breathing rate or air volume from trace or table

Numeric work on breathing: breaths per minute, volume differences, minute volume and oxygen used.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-BREATHING-RATE-VOLUME-CALC

1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

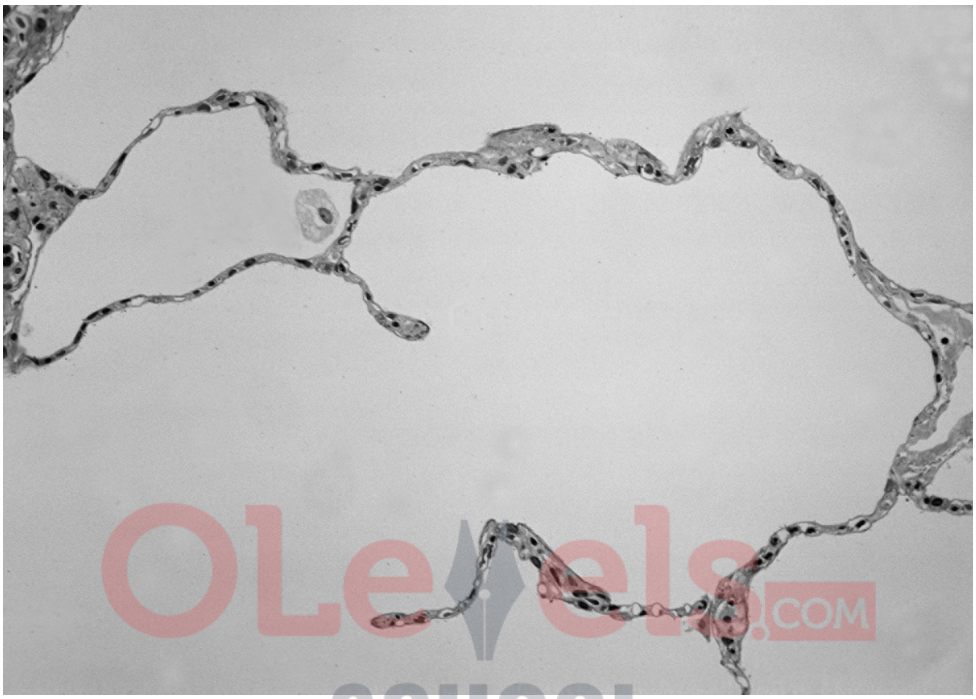
Worked-example retry · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

.....

2 .....

.....

1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part b.iii only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(iv) Explain the effect of increasing the percentage of carbon dioxide in the air a person breathes in on the tidal volume and rate of breathing.

(4)

(Total for Question 2 = 14 marks)

OLevels.com SCHOOL

Working and answer

2. Question 2 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

2 Aerobic respiration is the process that releases energy for use in the body.

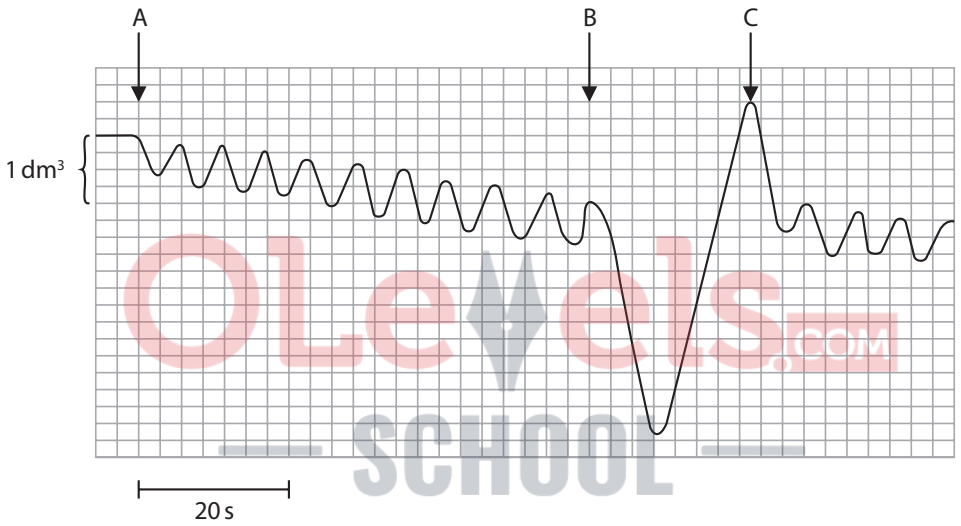
(a) Give the word equation for the process of aerobic respiration.

(4)

..... + ... → ..... + ... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.



(i) Calculate the rate of breathing between points A and B shown on the trace.

(2)

rate = ... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

(3)

difference in volume = ... dm^3

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 6 4 3 3 A 0 4 2 4

2. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 2 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

2 Aerobic respiration is the process that releases energy for use in the body.

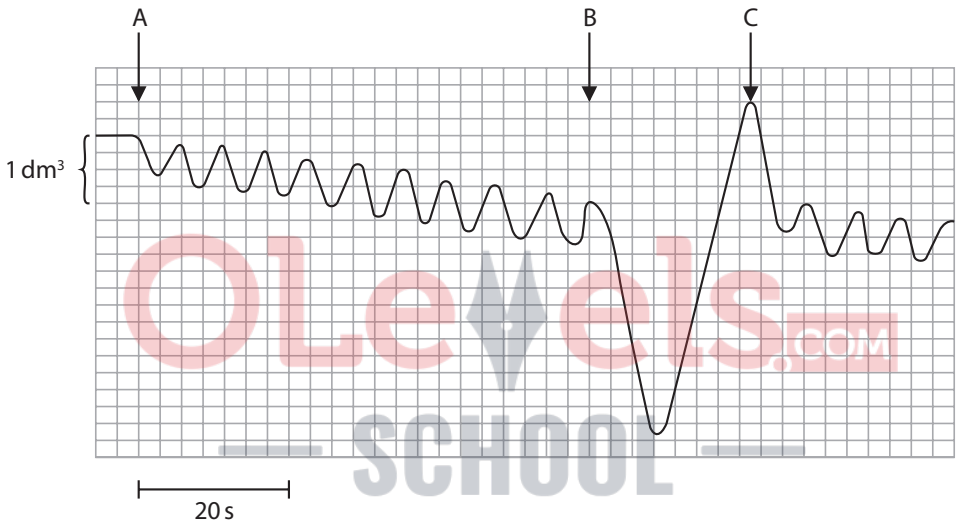
(a) Give the word equation for the process of aerobic respiration.

(4)

..... + ... → ..... + ... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.



(i) Calculate the rate of breathing between points A and B shown on the trace.

(2)

rate = ... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

(3)

difference in volume = ... dm^3

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 2 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

(c) Describe the differences that would be seen in the trace between points A and B if the person had been exercising vigorously before the readings were taken.

(2)

.....

.....

.....

.....

(Total for Question 2 = 11 marks)

Working and answer

OLevels.COM SCHOOL

4. Question 2 · part b.ii · 2 marks

June 2023 Paper 1

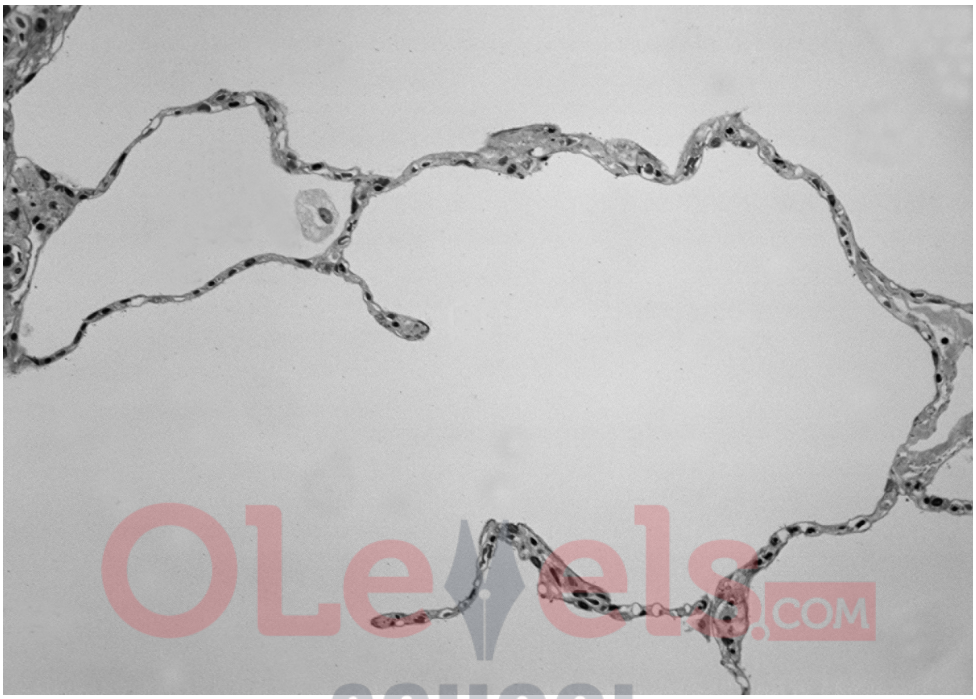
Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

.....

2 .....

.....

4. Question 2 · part b.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 5 8 5 A 0 4 2 0

4. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL